

Khizar Ali

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Professional Summary

AI undergraduate specializing in Computer Vision and Deep Learning. Experienced in building end-to-end ML systems including model training, optimization, deployment, and web/GUI integration. Strong foundation in neural networks, data processing, and scalable AI development.

Education

FAST-NUCES, Peshawar

B.S. Artificial Intelligence

Aug 2022 – June 2026

GPA: 3.5 / 4.00

Technical Skills

Languages: Python, C, C++, Dart

ML/AI: TensorFlow, Keras, Scikit-learn, CNN, VGG16, VGG19, MobileNet

Computer Vision: OpenCV, Image Processing, Real-time Detection

Web/Deployment: Flask, HTML, CSS, JavaScript, Streamlit

Data & Tools: NumPy, Pandas, MySQL, Git, GitHub, Hadoop, Spark

Experience

AI Engineer – NuSYS Research Lab, FAST-NUCES

Aug 2025 – Sep 2025

- Built Computer Vision module for multimodal psychology bot.
- Integrated OpenCV and Google ML Kit in Flutter for real-time processing.
- Contributed to system-level AI architecture design.

Teaching Assistant – FAST-NUCES

Jan 2025 – May 2025

- Assisted in AI course delivery, labs, and model debugging.

AI Intern – Sync-Intern (Remote)

Sep 2023 – Oct 2023

- Developed plant disease detection GUI using VGG19.
- Implemented neural network for handwritten digit recognition.

Projects

Pneumonia Detection (CNN + VGG16) Built deep learning pipeline for chest X-ray classification with Flask-based deployment.

Real-Time Facial Emotion Detection (MobileNet + OpenCV) Implemented live emotion recognition system with Streamlit interface.

Plant Disease Detection (VGG19 + Tkinter) Designed CNN-based classifier with desktop GUI integration.

Final Year Project

FetoVision (Medical AI / Computer Vision)

- Built an AI-based system for automated fetal biometry estimation from ultrasound images
- Developed U-Net-based segmentation models to detect fetal structures (head, abdomen, femur)
- Implemented measurement extraction for HC, AC, and FL using computer vision techniques
- Integrated Explainable AI (Grad-CAM) for model interpretability
- Designed an end-to-end pipeline supporting ultrasound image/video analysis with reduced manual effort

Core Competencies

Computer Vision • Deep Learning • Model Optimization • Neural Networks • Data Preprocessing • AI Deployment • Multimodal Systems • Research-Oriented Development